Search Frictions and Welfare in a Market for Prominence

Theory and Experiment

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Market for Search Prominence

Search prominence: Sellers pay platforms to make their products **easier to discover and inspect** for consumers.

- Examples: top advertising slot (online), city-center restaurant location (physical).

Costly consumer search: Searching for a good fit takes time and effort.

Value of prominence depends on consumers' search and purchase decisions.

This paper

A model of competition for prominence followed by pricing and consumer search in the product market.

An experiment with exogenous variation in search cost: Low vs. High.

Research Question

How does **search friction** affect the **level and distribution of surplus** in a market for prominence through strategic interactions among sellers and consumers?

Environment

2 sellers: bid for prominence, then set prices in the product market.

1 consumer: unit demand, observes both prices, searches to learn match values.

Product differentiation: Match values are i.i.d. across positions.

$$v_k = \begin{cases} V & \text{with probability } q \\ 0 & \text{with probability } 1-q \end{cases} \qquad k = 1 \text{ (top)},\ 2 \text{ (second)}$$

Roadmap

1. Model
2. Experimental Design
3. Results: Welfare and Behavior
4. Discussion

Timing of the Game

Auction

Two sellers bid in a **second-price auction** for **a top position**. (Bidding)
Winner = position 1, pays losing bid $b^{(2)}$.
Loser = position 2, pays 0.

Pricing

Given their positions, sellers **simultaneously** choose prices (p_1, p_2). (Pricing)
Sellers do not observe v_1 or v_2.

Search

Consumer observes (p_1, p_2, v_1) **for free**.
May **pay $s > 0$** to reveal v_2. (Search)
Buy from any **searched** seller (free recall) or exit (outside option 0). (Purchase)

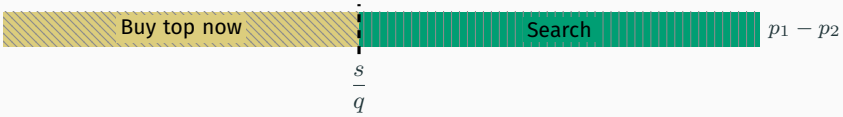
Seller profit = price charged $\cdot \mathbb{1}_{\text{sale}}$ − losing bid $\cdot \mathbb{1}_{\text{win auction}}$

Consumer utility = (match value − price paid) $\cdot \mathbb{1}_{\text{purchase}} - s \cdot \mathbb{1}_{\text{search}}$

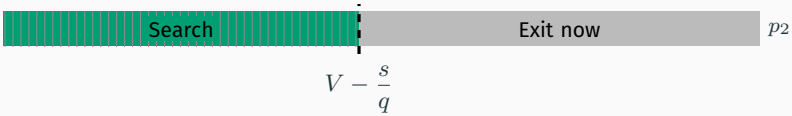
Search Cutoffs Depend on Price(s)

Observing (v_1, p_1, p_2), the consumer compares the expected gain from search with its cost s.

- If $v_1 = V$: The consumer searches iff $q\,(p_1 - p_2) > s \iff \boldsymbol{p_1 - p_2 > \frac{s}{q}}$

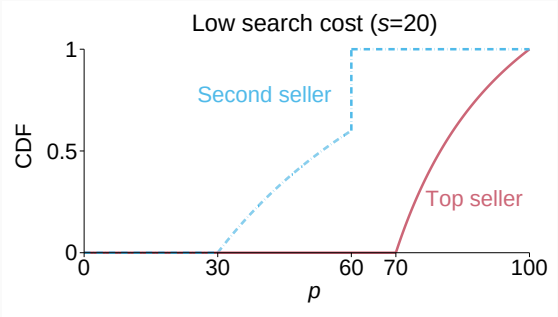


- If $v_1 = 0$: The consumer searches iff $q\,(V - p_2) \geq s \iff p_2 \leq V - \frac{s}{q}$



Price-directed search: Prices affect whether the consumer searches further. Search Path

Mixed Strategies in the Pricing Subgame ($V = 100$, $q = 0.5$)





Predictions

Price dispersion: Both sellers mix over disjoint price intervals.

Price hierarchy: $p_1 > p_2$. Top seller exploits **market power** from its search-cost advantage.

Price gap: $p_1 - p_2$ increases in s (comparative statics).

Equilibrium Bids for the Prominent Position

Expected revenue (continuation value) at each position:

$$\Pi_1 = q[(1-q)V + s], \qquad \Pi_2 = q(1-q)\left(V - \frac{s}{q}\right)$$

Equilibrium willingness to pay for prominence (bids) equals its **incremental revenue**:

$$b^* = \Pi_1 - \Pi_2 = s$$

Both sellers submit the same equilibrium bid.

Profits are equalized across positions:

$$\underbrace{\Pi_1 - b^*}_{\pi_1} = \underbrace{\Pi_2}_{\pi_2}$$

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Experimental Design

Between-subjects at UVA Veconlab

Low search cost ($s = 20$): 3 sessions, 42 subjects.
High search cost ($s = 40$): 3 sessions, 45 subjects.

Each market: 2 sellers + 1 consumer.

- 30 rounds with fixed roles and random rematching.
- Match values (v_1, v_2) independently redrawn each round.

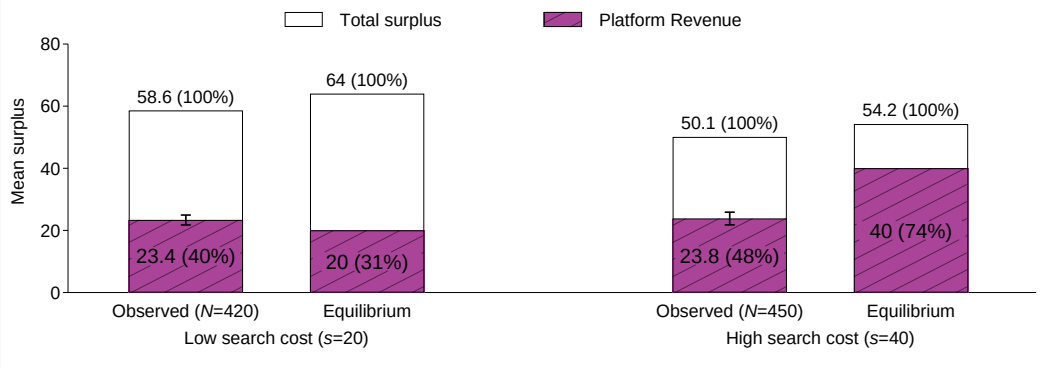
Feedback: Market outcomes after each round (v_2 only if searched). Feedback

Payment: $10 show-up fee + one randomly selected round; average $31.96.
Initial balance of 200 points (worth $20), reset each round to avoid losses.

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Welfare Result 1: Platform Revenue

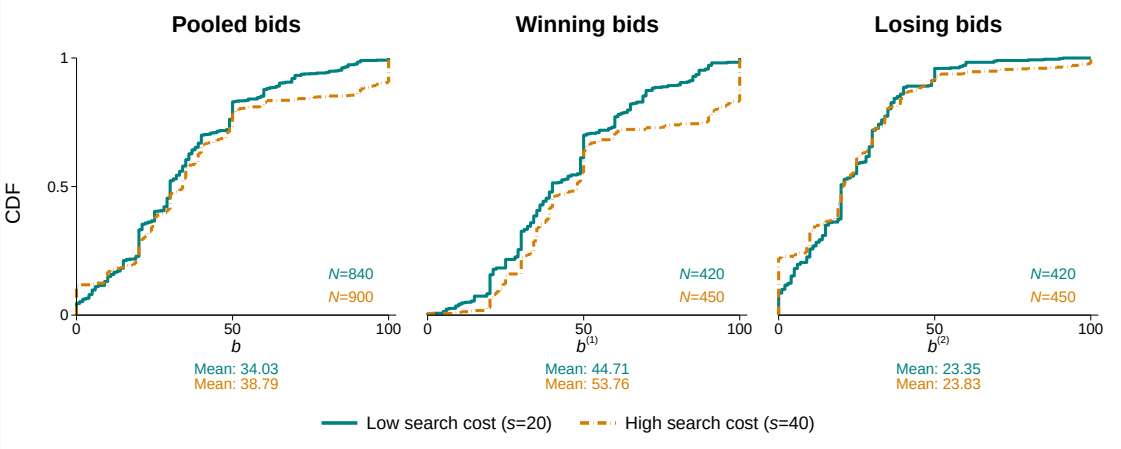


Equilibrium PR **doubles** as search cost doubles.

Observed PR **does not differ** significantly as search cost doubles ($p = 0.718$).

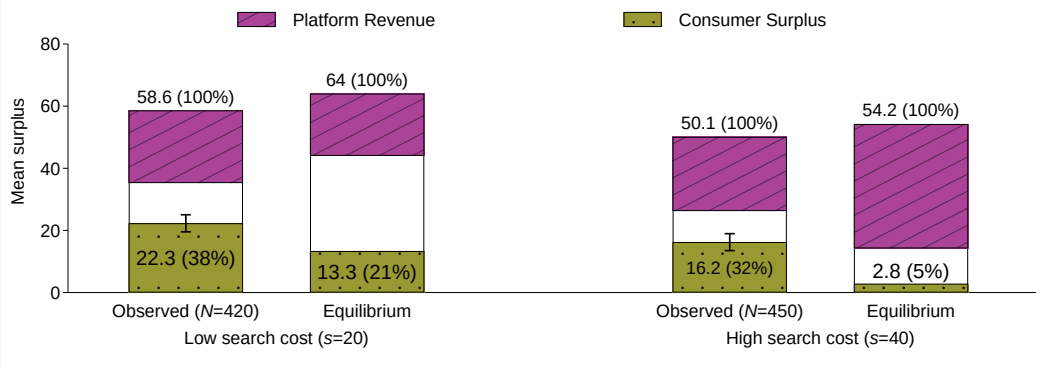
Observed PR is **above** equilibrium at $s = 20$ and **below** at $s = 40$ (both $p < 0.001$).

Explanation: Insensitivity of Bids



Insensitivity of (losing) bids ⇒ insensitivity of platform revenue.

Welfare Result 2: Consumer Surplus

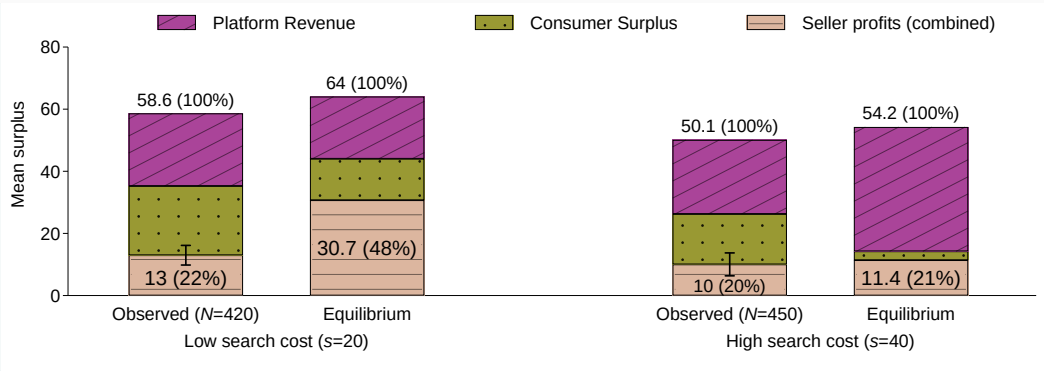


Equilibrium CS **decreases** by 78.7% as search cost doubles.

Consumers receive **more surplus** and a **larger share** than predicted at both search costs.

As search cost doubles, CS falls by **27.2%**, much less than the predicted **78.7%** ($p < 0.001$).

Welfare Result 3: Seller Profits



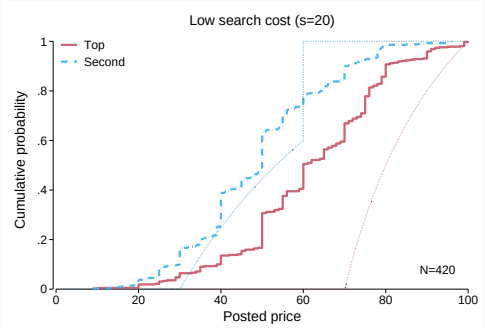
Equilibrium seller profits account for **48%** of total surplus at $s = 20$ and **21%** at $s = 40$.

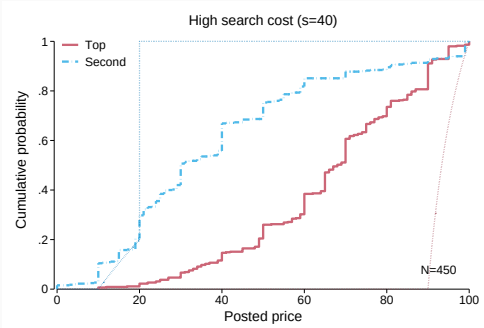
At $s = 20$, profits are **below** equilibrium ($p < 0.001$).

At $s = 40$, profits **do not differ significantly** from equilibrium ($p = 0.358$).

Sellers jointly retain **20–22% of total surplus** after paying for prominence.

Mispricing Relative to Equilibrium





Over-dispersion: p_1 and p_2 are more dispersed than predicted at both search costs.

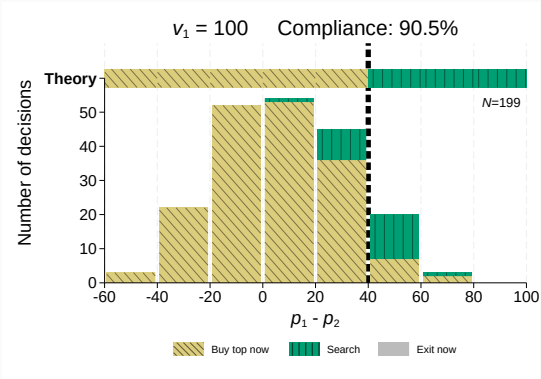
Price hierarchy holds: $p_1 > p_2$ at both search costs (one-sided KS test $p < 0.001$).

Lower top prices: $p_1 < p_1^{\text{eqm}}$ at both search costs (one-sided KS test $p < 0.001$).

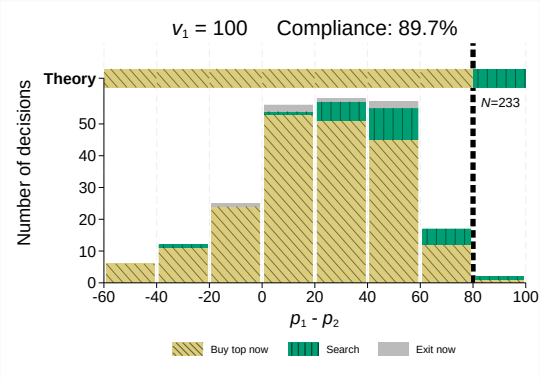
Compressed price gap: $p_1 - p_2 < p_1^{\text{eqm}} - p_2^{\text{eqm}}$ at both search costs (one-sided KS test $p < 0.001$).

Consumer Search: Good Top Match

Low search cost ($s = 20$)



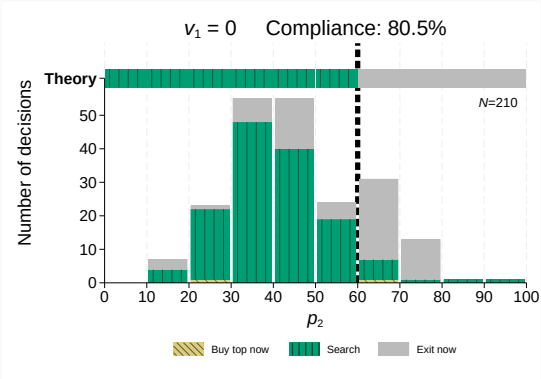
High search cost ($s = 40$)



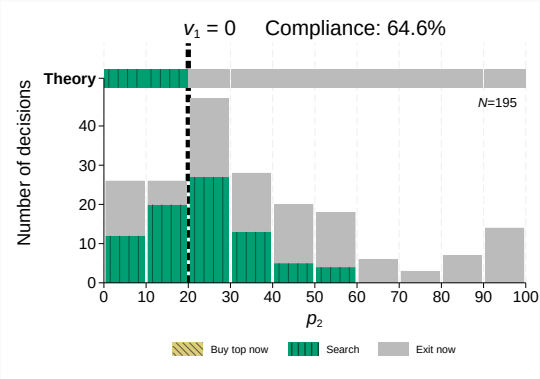
Consumers generally follow the search cutoff at both search costs.

Consumer Search: Bad Top Match

Low search cost ($s = 20$)



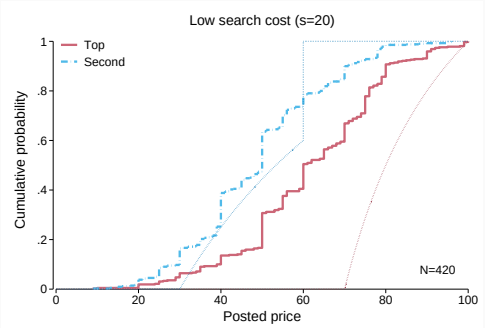
High search cost ($s = 40$)

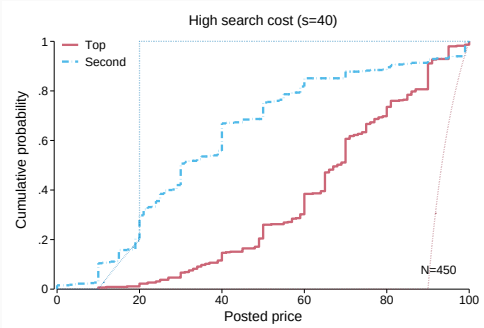


Search follows the cutoff, with more errors at the high search cost.

Search errors reduce CS $\Rightarrow$ cannot explain CS above eqm.

Explanation: Mispricing Benefits Consumers but Hurts Sellers





Compressed price gap $(p_1 - p_2) \Rightarrow$ more immediate purchases from top by $v_1 = 100$ consumers.

Lower top prices $(p_1) \Rightarrow$ higher CS and lower seller revenue per transaction at top.

Over-dispersion $\Rightarrow p_2$ above the search cutoff $\Rightarrow$ more exit by $v_1 = 0$ consumers $\Rightarrow$ lost sales.

Agent Quantal Response Equilibrium

(McKelvey and Palfrey, 1998)

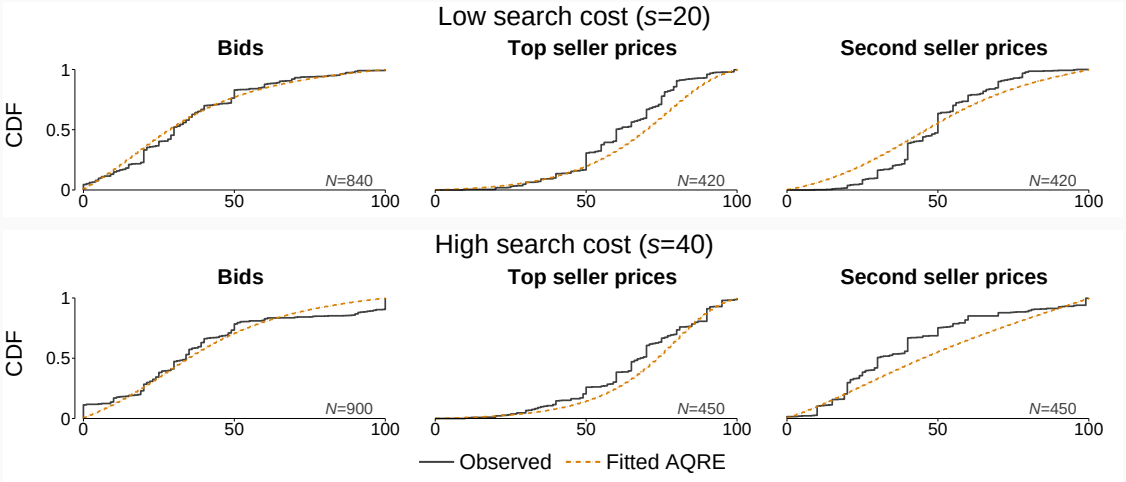
At each information set, an agent uses a **better response** rather than a best response. Choice probability $\propto \exp\{\lambda \times$ expected continuation payoff$\}$

Each agent **anticipates noise** in other players' and its own player's future decisions.

One common precision λ is estimated across all decisions and both search costs.

Agent Quantal Response Equilibrium: Fitted Bids and Prices

$$\widehat{\lambda} = 0.1146 \text{ (SE} = 0.0101), \qquad \widehat{\lambda} > 0 \ (p < 0.001)$$



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Summary

- A model relating the value of prominence to pricing and search behavior.
- Experimental variation in search cost.
- Insensitive bids and mispricing explain the distribution of surplus, while search behavior is close to optimal on average.
- AQRE captures noisy bidding and pricing behavior well.

Discussion

1. Do firms (sellers) actually make **mistakes**? After learning?

2. **Asymmetric quality** ($q_1 \neq q_2$): price signaling and auction sorting.

3. **Sponsored search auction** with endogenous consumer search: multiple tiers of prominence and many sellers and buyers.

4. **Platform** incentives to manipulate the search environment.
   - Optimal choice of search cost depends on revenue composition.
   - Personalized recommendations and demand signals to sellers.

Thank you!

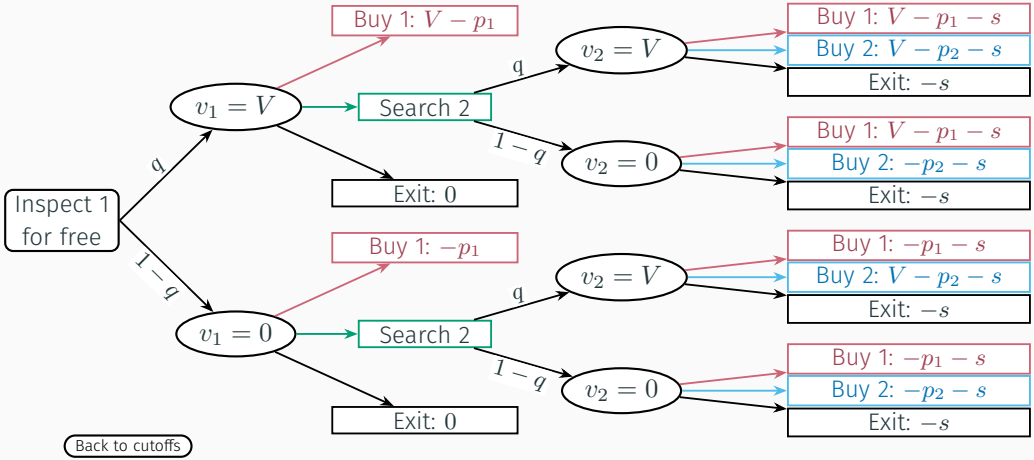
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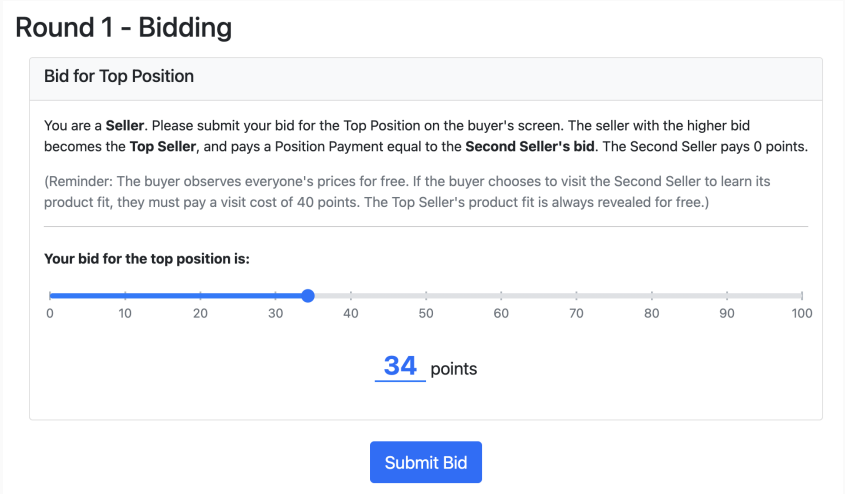
economics.virginia.edu/veconlab

Consumer Search Path



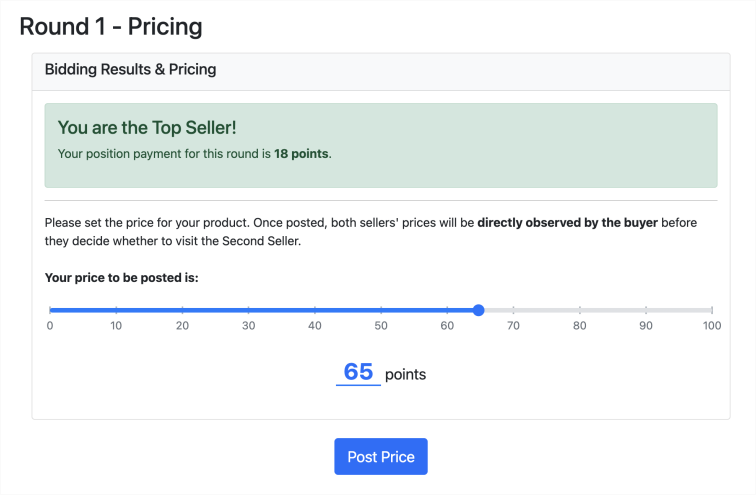
Back to cutoffs

Sample Bidding Screen



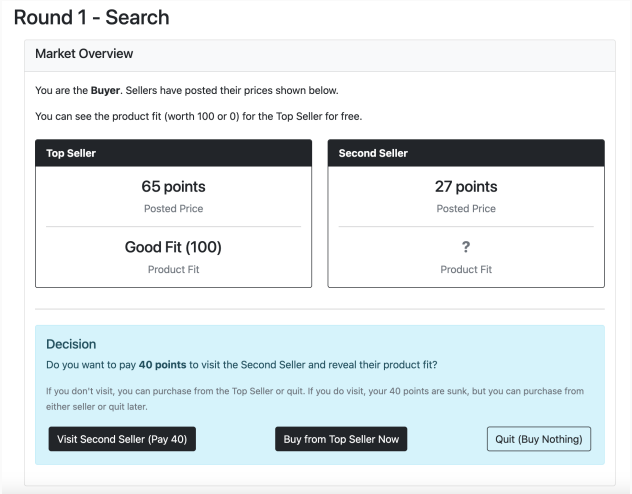
Back to game

Sample Price-Setting Screen



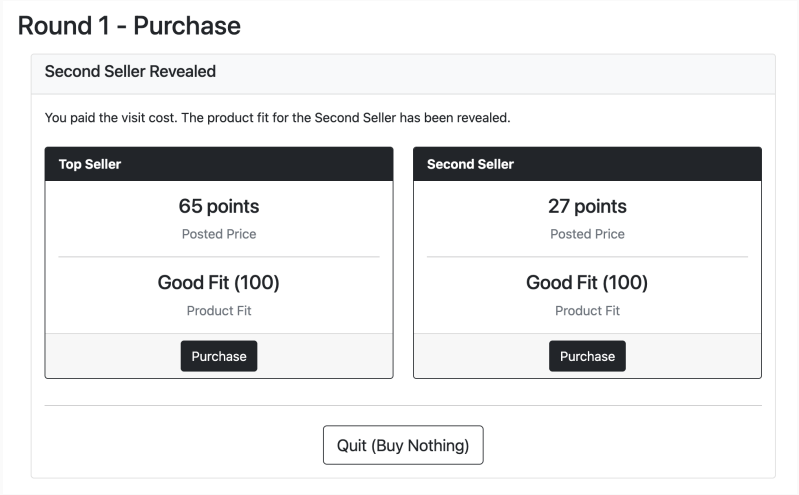
Back to game

Sample Screen for the Search Decision



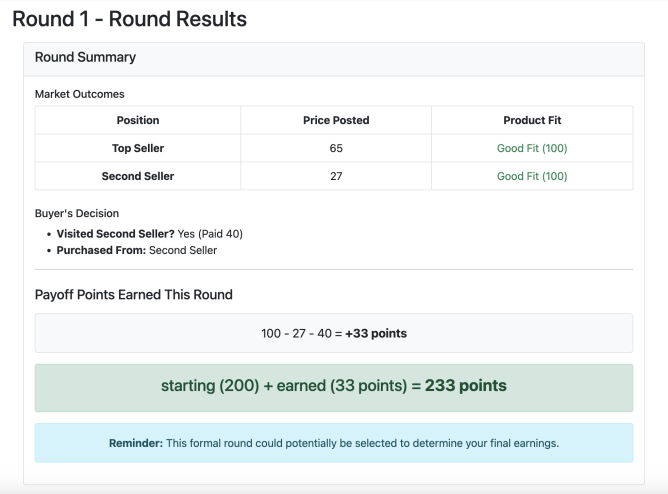
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Sample Screen for Purchase After Search



Back to game

Sample Screen for Round-End Feedback



Back to design